Thermal analysis of SMA,SMB, SMC packages

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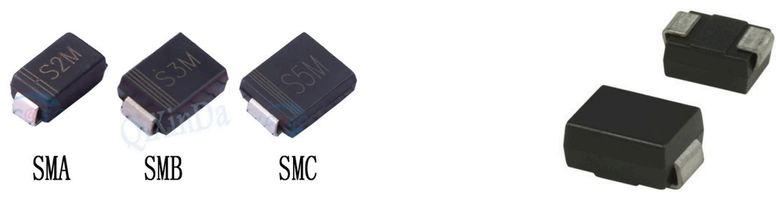
What are SMA, SMB, SMC packages?

- SMA,SMB, SMC refer to standard sizes of Surface Mount Device (SMD) packages, specifically for diode like TVS(transient voltage suppression) diode
- The primary difference between these packages is their physical size and power handling. As the package gets larger, it can dissipate more heat and handle more current.

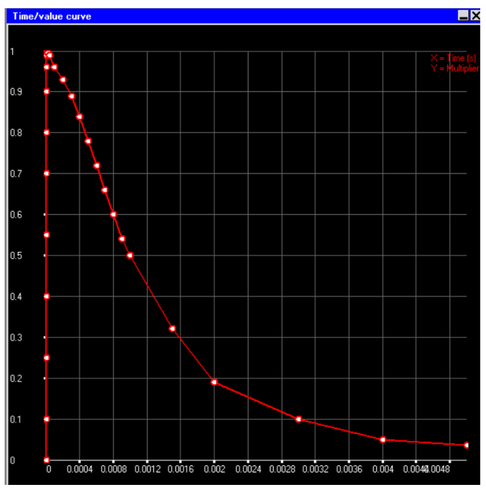


Size: SMC > SMB >SMA

Objectives and Methodology

- Analysis of a package for different sizes of cuboidal silicon die..
- Than comparison of cuboidal die with cylindrical die.
- Overall comparison of SMA, SMB, SMC packages.

- Modeled the packages on ansys icepak with the help of blocks.
- Applied power of 600W (10/1000 us) on the top of Silicon Die (not covering the full surface) for all the packages.
- Than did the transient analysis and got the temperature vs time plots and contour plots.
- Package stack: Silicon die → Die attach (0.03 mm) → Lead frame → Solder → PCB

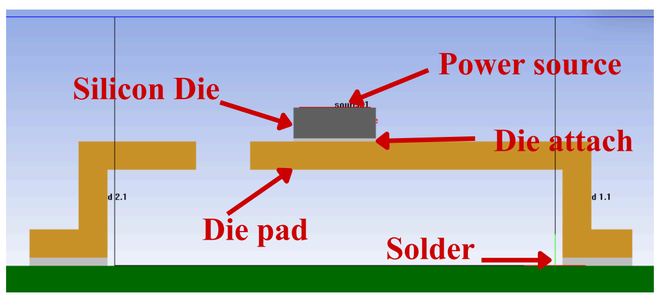


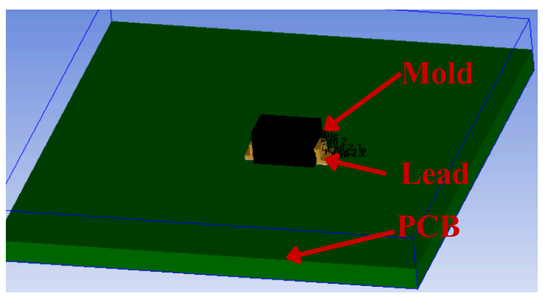
Power waveform (10/1000us)

Size details:

- Size of SMA: 4.37×2.615×2.095 mm
- Size of SMB: 4.315×3.62×2.285 mm
- Size of SMC: 6.855×5.905×2.31 mm
- We used different silicon die sizes:
  0.8×0.8×0.3 mm
  1.2×1.2×0.3 mm
  1.4×1.4×0.3 mm
  1.65×1.65×0.3 mm
- For shape we used cylindrical die and compared it with cuboidal die (1.2×1.2×0.3mm) by using same surface area as cuboidal.

Model view



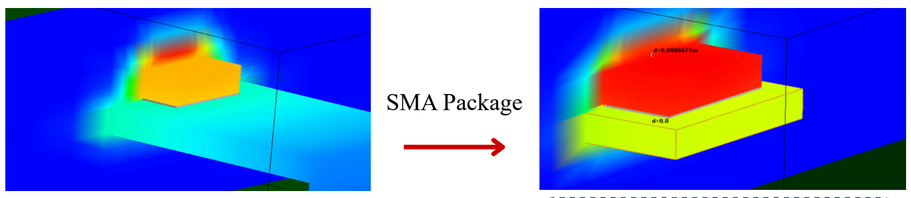


Material	Conductivity K	Relative Conductivity	Thermal Capacity CP	Density	Volumetric Heat Capacity	Relative Volumetric Heat Capacity
	(W/m K)	–	(J/kg K)	(kg/m3)	(J/K m3)	–
Epoxy Mold Compound	0.72	0.00200	794	2020	1603880	0.4748
Silicon (at 25C)	148	0.41111	712	2328.9	1658177	0.4908
SnPb Solder	50	0.13889	150	8500	1275000	0.3774
Silver Filled Die Attach	2.09	0.00581	714	3560	2541840	0.7524
CU Lead Frame	360	1.00000	380	8890	3378200	1.0000
FR-4	0.35	0.00097	878.6	1938	1702727	0.5040
Air	0.03	0.00008	1007	1.16	1170	0.0003

Role in package

Silicon die: Heat source/junction
Solder: Package to PCB bonding
Leadframe(Cu): Primary heat flow path
Mold compound: Lowest K (limits dissipation)
Die Attach : Intermediate bonding layer

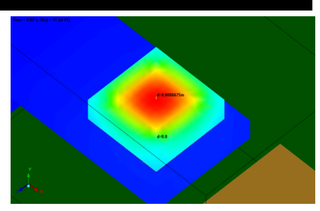
Temperature Contours plots



Die size: 0.8mm x 0.8mm x 0.3mm

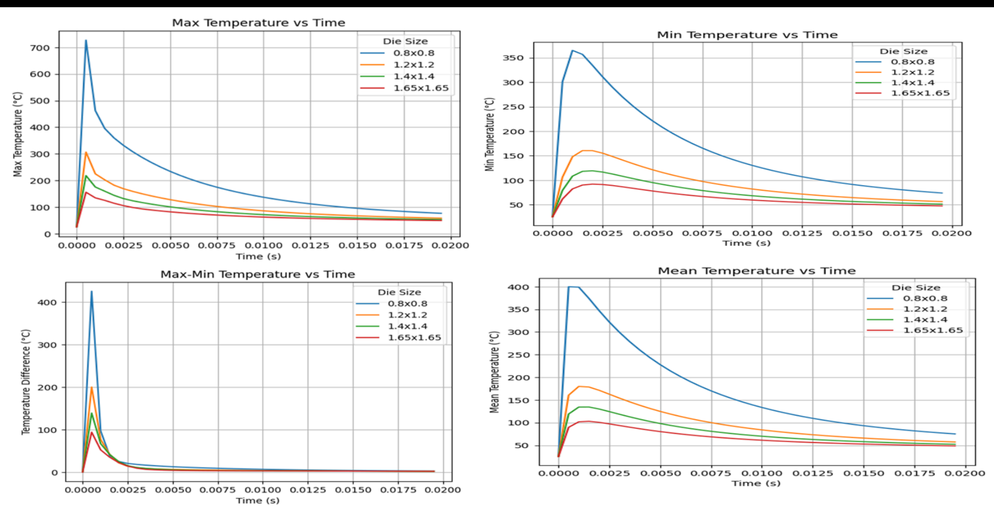
Die size: 1.2mm x 1.2mm x 0.3mm

Upon increasing the die size, the heat is transferred more effectively and evenly. The temperature is high for a greater distance on a smaller die, resulting in more breakdown of the mold for long-term operation. The die pad/lead heats up more quickly on the larger silicon die due to increased surface area.



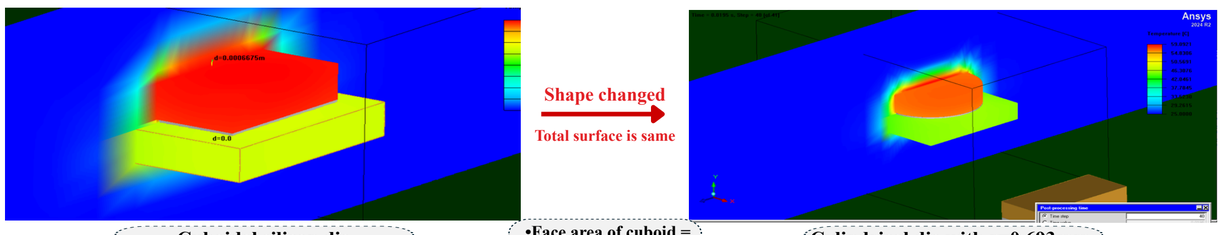
A hotspot started generating on the top of the die because the power is not applied to whole surface area. Around 70% area is in contact with the source.

Plots for different Die sizes



Increasing the die size fundamentally improves the thermal performance of the packages by increasing the thermal mass to absorb heat spikes and expanding the surface area to lower thermal resistance. Consequently, larger dies run cooler and maintain better temperature uniformity under the same power loads.

Comparison with Cylindrical die

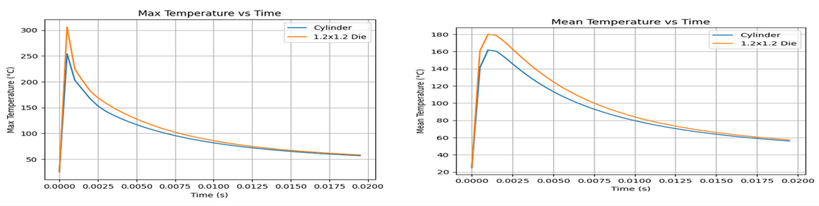


Cuboidal silicon die 1.2mm × 1.2mm × 0.3mm

•Face area of cuboid = 1.44mm^2
•Face area of cylinder = 1.49mm^2

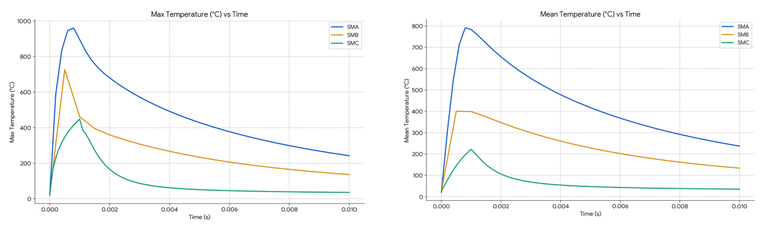
Cylindrical die with r=0.693mm, h=0.3mm

•Heat is transferred more uniformly on the cylindrical die compared to the cuboidal die due to the absence of corners.
•Additionally, since the cylinder has more face area than the cuboid which is in contact with lead frame, more heat is flowing from the cylinder to lead frame resulting in lower die temperatures.



As mentioned previously due to more face area of cylinder the peak temperature are less for the cylinder, as you can observe in this plots.

Overall comparison of packages



Plots for 0.8x0.8x0.3mm cuboidal die

SMC is the best at thermal performance. Its larger physical footprint and larger leadframe/thermal pads provide much lower thermal resistance, allowing it to efficiently handle the heavy 600W load. **SMB** sits in the middle but still reaches temperatures that would severely degrade or destroy standard silicon in a real-world scenario if sustained. Meanwhile, **SMA** performed the worst in this specific dataset configuration, suffering from extreme localized heating and exceptionally poor cooling rates.

Conclusion

- **As we increases die size in a package the overall thermal performance increases, max and mean temperature decreases decreasing the chance of wear and tear of the package.**
- **The cylindrical die thermal performance is better than the cuboidal die. Due to more uniform heat distribution and lower max temperature.**
- **Between SMA,SMB and SMC, SMC is the best at thermal performance due to the reason stated above.**

Future work

- **Calculate quantitative thermal resistances for each configuration**
- **Determine maximum safe power supply to avoid PCB solder melting**
- **Extend analysis to 4000W at 8/20 µs waveform**
- **Investigate forced convection cooling, heat sinks, and thermal vias in PCB**
- **Explore alternative die attach materials, solders, and mold compounds with higher thermal conductivity**

Acknowledgement

Texas Instruments

References

[1] Texas Instruments (D. Edwards, H. Nguyen), "Semiconductor and IC Package Thermal Metrics," Application Note SPRA953D, 2024. [2] Freescale Semiconductor (NXP), "Thermal Analysis of Semiconductor Systems," White Paper BASICTHERMALWP, Rev. 0, 2008.[3] Vishay Semiconductors, "Outline Dimensions: SMA-1 (DO-214AC)," Document No. 97292, Feb. 2025. [4] Freescale Semiconductor (NXP), "Thermal Analysis of Semiconductor Systems," White Paper BASICTHERMALWP, Rev. 0, 2008.